The sport of figure skating sits in a unique and often controversial place in the world of competitive sports. Unlike the likes of swimming or track and field, where a participant is declared victorious at the concrete decision of a stopwatch or similar measure, figure skating relies entirely on the subjective evaluation of a human judging panel. In recent times, this inherent subjectivity has reached multiple levels of controversy, most notable in the 2002 Salt Lake City Winter Olympic Games. Here, a vote trading scandal in the pairs event forced the International Skating Union (ISU) to entirely scrap their traditional 6.0 system that was the staple for many decades before, in favor of a new, less controversial system.

This new system, the International Judging System (IJS), had its foundations rooted in a mathematically rigorous framework designed to work to eliminate bias and increase transparency in judging. Under the IJS, all elements performed by the skater are assigned a standardized Base Value, coming from a publicly announced Scale of Values. Judges then modify this value using a Grade of Execution (GOE), based on their own subjective viewing of the element. Alongside this technical aspect, the judges award Program Component Scores (PCS) to demonstrate their evaluation of a skater's artistic and choreographic merits.

At the advent of this new, innovative system, the IJS was pronounced and heralded as the end of corruption in figure skating judges. However, more than two decades later since its inception, media narratives and commonplace discussion remain dominated by accusations of corruption, outrage against perceived judging bias, and now more than ever, disagreement drawn along strict political lines. This constant media outrage raises a critical concern: is national judging bias an actual systemic epidemic within the sport and the IJS, or simply a byproduct of what is marketable to such a niche and passionate fanbase?

With this paper, I seek to answer that question through the visualization of both frequency and magnitude of national judging bias, specifically focusing on the ISU World Figure Skating Championships. In hopes of providing the most comprehensive analysis possible, while balancing the understanding of the lay casual figure skating enjoyer, this paper utilizes a tiered scaled approach. First, it examines the micro scale, isolating specific instances of extreme individual bias from a skater's nation's judge, whilst providing guidance on understanding the

underlying technicalities of the IJS. Next, it zooms out to the mega scale, evaluating how specific events of bias impact competitive standings at single events. There is always the question of how bias affects placements. Finally, it touches the macro scale, seeking to identify long term trends and variations within the sport's four disciplines. As there is data from more than a decade of international competition, it will examine the overarching effect that the IJS has had for the sport. By mapping the data and performing critical analysis, I attempt to look past journalistic embellishment to reveal what else, aside from scandal, we can glean from a decade of protocols.

To investigate whether there is a presence of national bias in the sport of figure skating, this paper analyses protocols published by the ISU for their World Figure Skating Championships event. This event, the pinnacle event of the sport, occurs annually and highlights the talents of the season's best skaters. These protocols encompass all four major disciplines: Men's Singles, Women's Singles, Pairs, and Ice Dance. They also capture scores for both the short and long segments of competition.

The time span for the dataset is from 2011 to 2025. While the IJS was introduced in 2004 directly following the 2002 Olympic scandal, it took many years of tinkering to get it to where it is now. While there still have been many changes since 2011, protocol sheets as we know them today have been published since then. These protocols are publicly available through International Skating Union announcements. However, because these protocols are published as complex, multi-layered PDF documents designed with the sole purpose of human readability, gathering and producing the dataset is not an easy task. Most notably, these documents are not designed for data extraction and analysis, and are formatted for the aesthetics of the host nation. As a result, littered throughout the documents are images, stray text, and overall inconsistencies throughout the years.

Using Claude AI by Anthropic, an automated parsing script was developed which iterated through the PDF protocols, scraping the tabular data into a much more structured format for easy interpretation and machine readability. However, even utilizing such a tool, the results were less than ideal. Manual editing was required, particularly when it came to double checking accuracy with judge names and skater rank/places.

As the data processing progressed, several unforeseen technical and structural hurdles mandated significant refinement of my methodology. First, through general exploration and internet research, I discovered that prior to the 2016-2017 season, the ISU had utilized a random draw system for judging panels. Instead of the current system with a transparent nine judge panel, twelve judges evaluated each event, while only nine were randomly selected by computerized means to count toward the official score for the skater. Additionally, these nine were anonymized per individual skater performance. This caveat to the protocols made it impossible to definitively map a specific home judge to a specific score for events. Further, the calculation of GOEs over the last decade proved even more difficult and complex than initially anticipated. Almost every year in the past decade has seen shifts in how GOEs are calculated. While most elements now use a percentage based scale, certain elements such as choreographic step sequences instead use fixed numerical values that change across different seasons. Additionally, percentage boosts that GOEs provided to the skaters' scores in past years were hardcoded into the Scale of Values, rather than being strictly proportional. To account for all these variations would require an immense amount of manual hardcoding.

To ensure the integrity and validity of the findings, I used SkatingScores.com to cross reference and fact check my parsed values. Also, given the sheet computational time required to verify elements across years of data, I narrowed the project's scope down to the top 15 skaters of each competitive segment. While this drastically reduced the amount of data this paper works with, it does help ensure that the analysis focuses on those skaters who are in medal and title contention. Significantly, those are the athletes whom the press is interested in, and where the stakes of national bias are higher. After these iterative cycles of parsing, verification, and scoping, the initial dataset was produced.

To understand how national bias can be quantified throughout the dataset, it is first necessary to understand how a single figure skating protocol is calculated under the IJS. An individual skater's performance score is the sum of two separate subscores: the Technical Element Score (TES) and Program Component Score (PCS). Every element, defined as a particular move performed by the skater, carries a predetermined Base Value (BV). Each judge on the panel then

evaluates the quality of that element and assigns a Grade of Execution (GOE). This GOE corresponds to a scaled numerical value that is added to or subtracted from the BV, and is currently based on percentage amounts. GOE currently ranges from +5 to -5, with +5 adding 50% of the BV onto the BV itself to get the final score for that element. Similarly, +4 adds 40%, while -2 removes 20%. The second subsection, the PCS, is to account for the performance side of the sport. Judges use standardized marks out of 10.0 for artistic and choreographical categories, most notably Composition, Presentation, and Skating Skills. These two scores combined make up a skater's Total Segment Score, or their score for that skate. In official ISU competitions, the highest and lowest GOE and PCS marks from the panel of judges are discarded, utilizing a trimmed mean to calculate the final panel score.

ISU World Figure Skating Championships 2017

MEN FREE SKATING JUDGES DETAILS PER SKATER

Rank	Name	Nation	Starting Number	Total Segment Score	Total Element Score	Total Program Component Score (factored)	Total Deductions
1	Yuzuru HANYU	JPN	19	223.20	126.12	97.08	0.00

#	Executed Elements	Info	Base Value		GOE	J1	J2	J3	J4	J5	J6	J7	J8	J9	Ref	Scores of Panel
1	4Lo		12.00		2.43	2	2	3	3	2	3	2	2	3		14.43
2	4S		10.50		2.71	3	2	3	3	2	3	2	3	3		13.21
3	FCCoSp4		3.50		0.86	1	2	2	2	1	2	2	2	1		4.36
4	StSq4		3.90		1.50	1	2	3	2	2	2	3	2	2		5.40
5	3F		5.30		1.50	2	1	2	2	3	3	2	2	2		6.80
6	4S+3T		16.28	x	2.43	2	2	3	3	1	3	2	3	2		18.71
7	4T		11.33	x	2.43	2	2	2	3	1	3	2	3	3		13.76
8	3A+2T		10.78	x	2.00	2	2	2	2	2	2	2	2	2		12.78
9	3A+1Lo+3S		14.74	x	2.14	3	2	3	3	1	2	2	1	2		16.88
10	FCSSp4		3.00		0.79	1	2	2	2	0	2	2	1	1		3.79
11	ChSq1		2.00		1.40	1	2	2	2	3	2	3	2	1		3.40
12	3Lz		6.60	x	1.50	1	2	2	2	3	3	2	2	2		8.10
13	CCoSp4		3.50		1.00	2	2	3	2	2	2	2	2	1		4.50
			103.43													**126.12**
	Program Components				Factor											
	Skating Skills				2.00	9.25	9.75	9.75	10.00	9.50	9.75	9.75	9.75	9.50		9.68
	Transitions				2.00	9.00	9.50	9.50	9.50	9.25	9.75	9.50	9.50	9.50		9.46
	Performance				2.00	10.00	10.00	9.75	10.00	9.25	10.00	9.75	10.00	9.50		9.86
	Composition				2.00	10.00	10.00	9.75	9.75	9.50	9.75	9.75	9.75	9.50		9.75
	Interpretation of the Music				2.00	9.00	9.75	10.00	10.00	9.25	10.00	10.00	9.75	9.75		9.79
	Judges Total Program Component Score (factored)															**97.08**

Deductions	**0.00**

x Credit for highlight distribution, base value multiplied by 1.1

Figure 1: A standard ISU judging protocol detailing the Technical Element Score (top) and Program Component Score (bottom) for Yuzuru Hanyu's Free Skating performance at the 2017 World Figure Skating Championships.

As seen in Figure 1, the harder, big ticket items such as quadruple jumps, hold more Base Value. In his program, Yuzuru Hanyu skated entirely clean, meaning no negative GOEs were awarded. As such, his final technical score is significantly higher than the total of his base values. Additionally, multiple judges awarded him 10.0s in his Performance and Composition categories, speaking towards their appreciation of his artistic talents.

Because the official protocols only provide final, trimmed panel consensus scores for the skater's performance, relying on merely the published totals obscures the specific behavior of the individual judges. However, we can see that each judge's GOEs awarded are present on the protocol. To isolate each judge's rulings, the parsed data was used to reverse-engineer and calculate individual judge scores. For every skater, the parsing script calculated exactly what the total would be if only a specific judge's column were counted. Bypassing the ISU's trimmed mean, the extreme high and lows awarded by individual judges are preserved.

On the protocols, judges are additionally anonymized as "J1," "J2," "J3," and so on. However, in the published protocols, there are tables for the panel of judges, providing information to map the judge assignments to the judge scores. Further, at ISU championship events, including the World Championships, judges' representative nationalities are anonymous as well, with all judges on paper representing the ISU. Luckily, at the beginning of each competitive season, the ISU announces the list of judges certified to judge during that season, including detailed information about the national federation each judge represents. Through the joining of the judge nationality data, judge panel data, and judge score data, the final dataset that this paper uses is attained. This allowed further for the creation of the core independent variable, home_judge, a boolean value detailing whether or not the nationality of the skater matched the judge. This enabled direct comparison between the score a skater received from their compatriot versus the scores awarded by the rest of the unaffiliated panel.

While this approach provides a clear baseline for identifying direct national bias, it is necessary here to acknowledge that there are indeed several methodological limitations and assumptions inherent in this paper and dataset.

First, treating national bias as a strict binary fails to capture the complex reality and relationship of geopolitical blocs. Historically, as seen in the 2002 Salt Lake City Winter Olympic Games, judges from politically allied nations have been known to artificially inflate the scores of each other's skaters. Additionally, bribery and judge coercion have also been seen as a different way of perpetuating bias. Because this analysis only flags direct on to one national matches, there is inherently a likely underestimation of the true scope of systematic bias.

Moreover, while the ISU generally reports final scores in this manner, combining GOE and PCS into a single combined segment score assumes that bias is applied uniformly across both subsections. It is highly plausible and frankly likely that judges find it easier to inflate an individual skater's PCS rather than their TES, as the PCS is generally regarded as the more subjective section. Future analysis and research could remedy this by disaggregating scores to determine for extreme cases of national bias, where on the protocol a home judge best hides their bias. However, for the purposes of this paper and generally establishing a baseline visual narrative, the combined segment scores serve as a robust indication of overall judging intent.

To begin, the most effective way to understand how national bias physically manifests on a skater's protocol is to isolate a single performance where a home judge was particularly biased on the panel. One such primary example can be found at the 2017 ISU World Championships Pairs Free Skating event, where the Russian Team Tarasova / Morozov were awarded significantly higher scores by their sole compatriot than the rest of the panel.

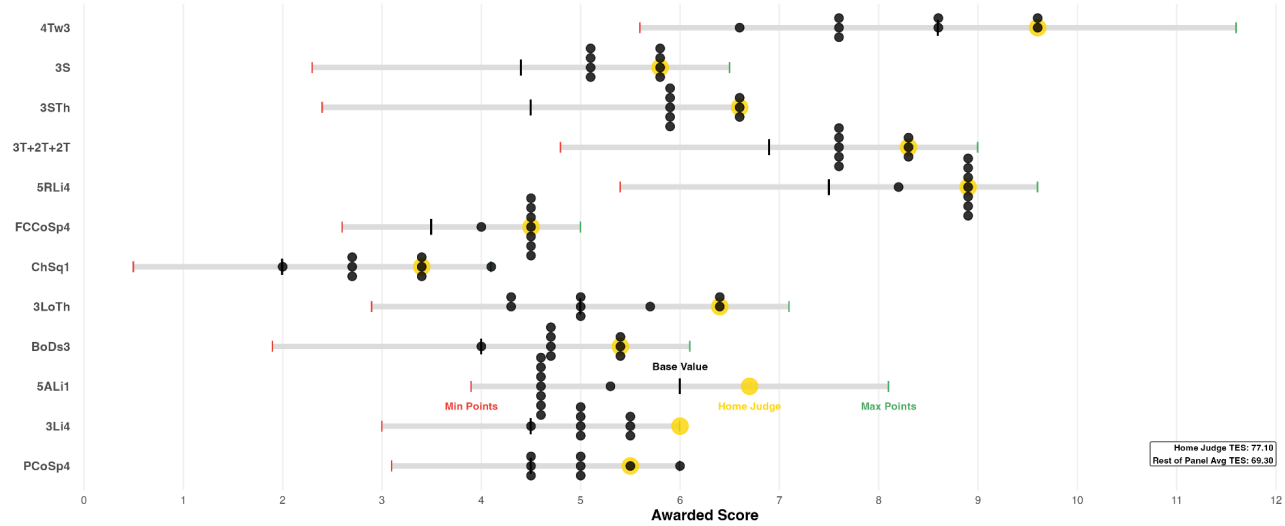


Figure 2: Distribution of awarded points for Tarasova / Morozov in the 2017 World Championships Pairs Free Skating. The home judge's scores (yellow) consistently deviate toward the maximum possible points compared to the rest of the panel (black).

Figure 2 illustrates the micro-scale manifestation of national bias, where I plotted the distribution of awarded points for Tarasova and Morozov in their 2017 Free Skate. The visualization clearly differentiates the Russian judge with the yellow, and the rest of the panel with the black. To increase comprehension, the error bars displayed note that range of points that the element performed could possibly attain through judge GOE. The aggregate impact of the Russian judge's scoring behavior is immediately apparent in the raw totals: the home judge awarded a Technical Element Score (TES) of 77.10, nearly eight points higher than the rest of the panel's average TES of 69.30. To put that into perspective, a cleanly landed quad jump might only receive eight points. Essentially, through only tinkering with the Grades of Execution, it is almost as if the team had performed a whole other event.

Looking element by element through this scatter distribution, this nearly eight point inflation was not the result of a few accidental mistakes, but rather a consistent, systemic elevation across the entire program. Across almost every technical element, the home judge's score anchors the extreme right hand limit of the distribution cloud. Most notably, for the Group 5 Axel Lasso Lift (5ALi1) and Group 3 Lift (3Li4), the home judge is completely detached from the rest of the panel's consensus. On these elements, the home judge awards a mark that approaches the

absolute maximum score that the element is able to attain, indeed reaching that number on the Group 3 Lift. For the Group 5 Axel Lasso Lift, every other judge on the panel deemed the element to have been flawed in execution, awarding it with negative GOE. However, the yellow dot is firmly to the right of the base value. Even on elements where the panel demonstrates tighter consensus, such as the Choreographic Sequence (ChSq1), the home judge consistently is near or at the top of that consensus.

This one distribution perfectly captures that vulnerability of the IJS and a phenomena we can call a safety net. Under the IJS trimmed mean, the extreme right and left leaning scores of the panel are discarded as the lowest low and highest high. However, by intentionally and consistently occupying that maximum outlier position, the home judge successfully protects the next highest score on the panel from being trimmed. By pushing the elements towards their absolute maximum score, the home judge artificially raises the scoring floor, mathematically insulting the skater's final technical total from the panel's harsher evaluations.

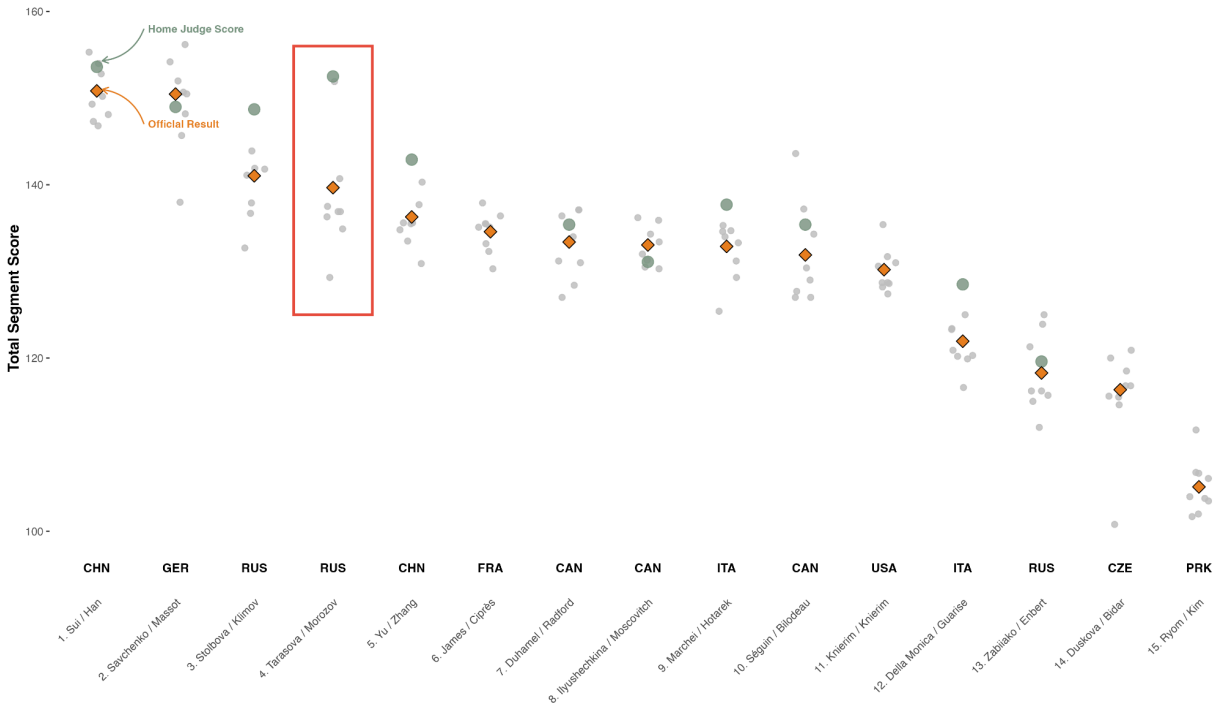


Figure 3: Total Segment Score distribution for the top 15 pairs at the 2017 World Championships Free Skating. The official trimmed consensus (orange diamond) is plotted against individual panel judge scores (gray dots). In instances where a judge is judging a skater from the nation

they represent, the score awarded by that judge is isolated (green circle), frequently appearing at the top of the scoring distribution.

While the micro-scale analysis proves that individual elements can be artificially inflated, Figure 3 zooms out to the mega-scale to demonstrate how these isolated bumps accumulate to impact overall event standings. Figure 3 plots the Total Segment Score distribution for the top 15 pairs at the 2017 World Championships so that we can visualize the full panel's variance for each skater's performance. It is very straightforward to compare the baseline orange diamond, the official score awarded, with the behavior of the compatriot home judge.

As this is the same instance discussed in Figure 2, we already know that the fourth place finishers in this event, Tarasova / Morozov, have the most outrageous variance in scores. In this chart, it is clearly shown that the green dot is miles higher than the general panel consensus. While the rest of the nudges awarded a score closer to the mid 130s and approaching 140, the Russian judge's dot sits squarely nearing 150. At this macro level, the mechanics of the safety net become even more clear. Because the home judge's score is a massive upper outlier, it is inevitably dropped for the final score. However, through this intentional absorption of the drop, it is guaranteed that the second highest score on the panel survives, dragging the orange diamond up.

Even more importantly than the boxed instance is that the trend continues for almost all skaters in this event. Looking across the field, the home judges for both the third place team (Stolbova / Klimov also from Russia), and the fifth place team (Yu / Zhang from China) similarly show the same trend. The extreme vertical spread seen in the middle of the pack is not a byproduct of judging difficulty, but instead a direct visual signal of national bias inflating the scores.

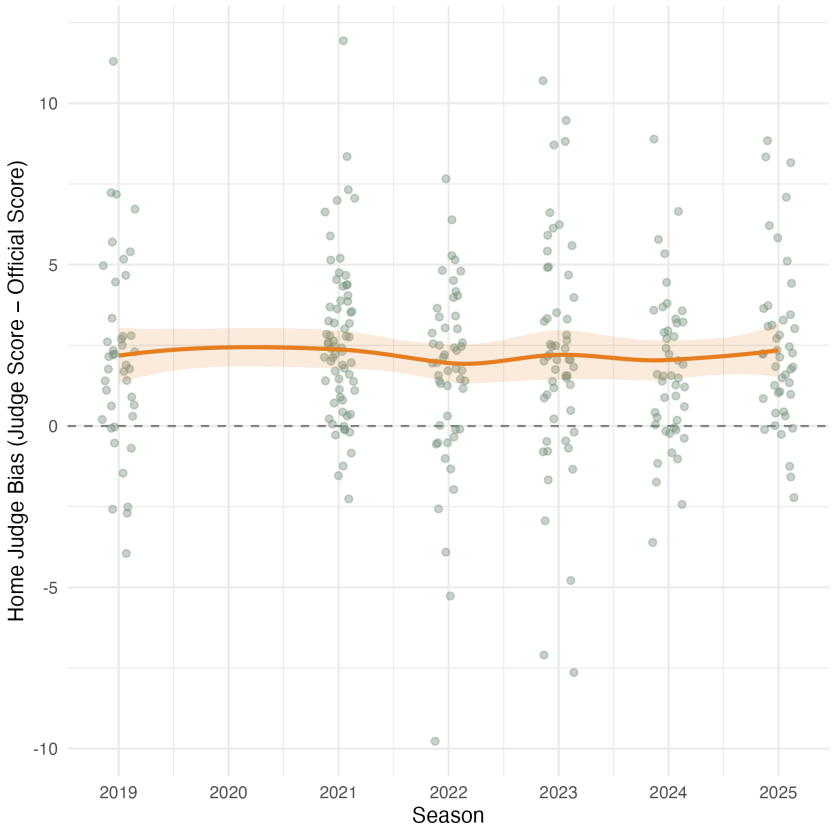


Figure 4: Average home judge segment score versus the official trimmed consensus, demonstrating a persistent parallel gap between compatriot inflation and panel averages. From 2019 to 2025, the trend remains relatively constant, with a consistent 2.5 boost given by home judges.

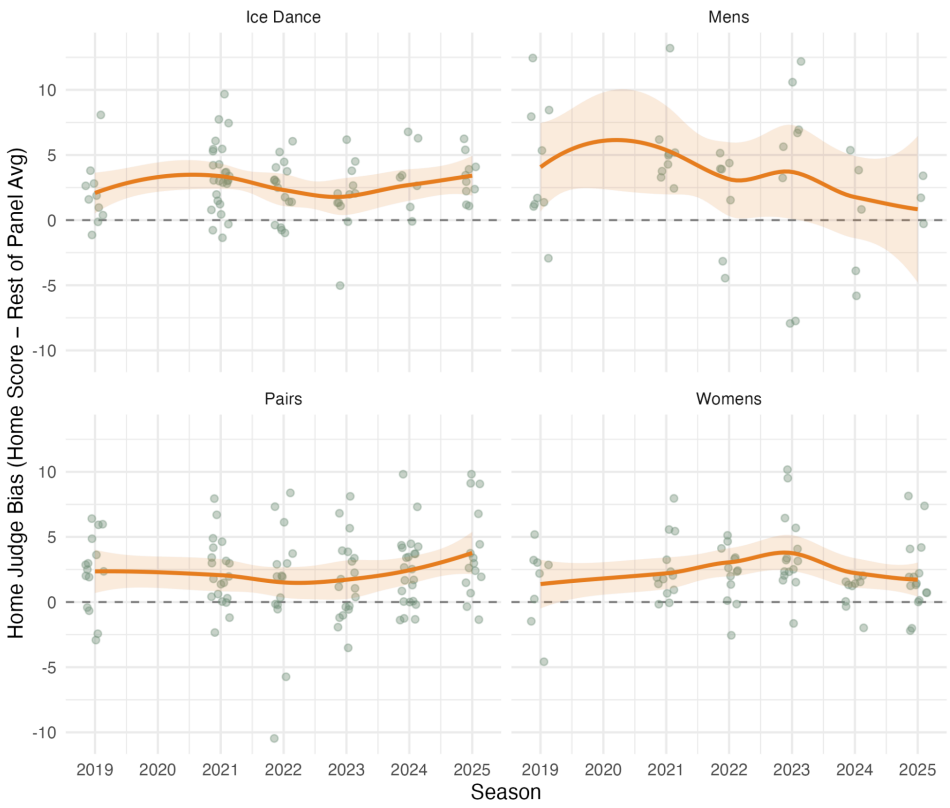


Figure 5: Average home judge point inflation broken down by discipline, with a smoothed line helping to show longitudinal trends. There are less data points for the Mens quadrant and much higher variation.

In order to determine whether the national bias observed at the micro and mega level can be observed throughout the history of the IJS, it is necessary to examine the data longitudinally. Figure 4 contextualizes the resiliency of this national bias across a seven year span, plotting the raw home judge segment score against the official trimmed consensus. We can see that throughout the modern era, the gap between the home judge and rest of the panel remains almost remarkably parallel. The persistent, two and half point margin between skaters' compatriots and the average of the rest of the panel is strikingly telling about the ingrained nature of this safety net effect. It is additionally, refreshing to know that there is no major trend upwards or downwards in the most recent years, which gives confidence in the IJS system.

Figure 5 details the average point inflation awarded by home judges faceted by discipline. The utilization of small multiples allows the comparison of how judges perform in different figure skating disciplines. Here, we see that rather than remaining stable or decreasing under supposed modern refinement, there is a bit of a slight increase in nationalistic scoring in recent years. It is important to note the Men's quadrant lacks the amount of data points, which explains why there is such more volatility with the smoother for that data. While further research analysis would most definitely prove helpful here, I can assume that this is due to more localized dominance of that event. The ISU is less likely to allow an American judge to keep judging the World Championships, if an American skater is the heavy favorite for years straight.

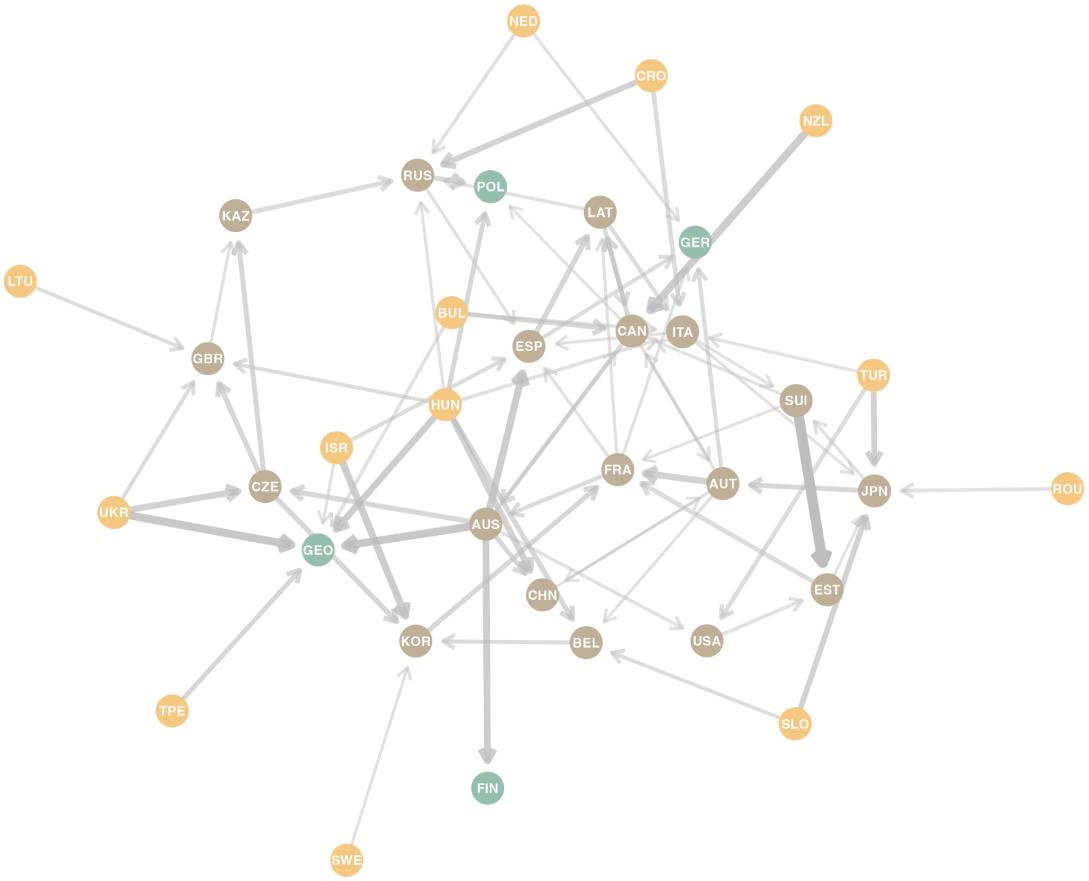


Figure 6: Network graph of systemic international judging bias. Arrows point from judging federation to skater federation, indicating when there is an average score inflation of over 1 point above the panel consensus. Line thickness correlates to the magnitude of the bias. Nodes are color coded by role with yellow indicating nations that only inflate others, green indicating

nations that only receive inflation, and the brown nodes indicating nations that engage in both directions.

While the micro, mega, and macro scales illustrate how national bias is executed in a single segment, across a single event, and throughout time, they cannot help us determine if these are merely isolated acts or symptoms of coordination between nations. To attempt to investigate the presence of bloc judging underneath the IJS, Figure 6 zooms out mapping any instance of significant (>1 point) of bias. By utilizing the "kk" layout for this network, the data is organized into visual clusters.

Peering closely at these visual clusters, while not the most specific, a compelling narrative can emerge regarding the reality of this bloc judging. The dense center of the network is heavily populated by the brown nodes, representing federations that actively participate in both inflating other nations' skaters and receiving inflated scores in return.This interconnectivity among specific groups strongly suggest a reciprocal dynamic among the sport's influential federations. While we will not know whether this is intentional or not, nonetheless it exists. Conversely, the periphery of the graph is additionally scattered with yellow and green notes, which more than not, seem to be the smaller nations. These federations either solely mark favorably or are favorably marked. While this network mapping is not the best at proving or determining collusion or intent, the mutually beneficial groupings visualized here provide at least some credence to rumors of the geopolitical alliances that the IJS was fundamentally designed to eradicate.

As this paper comes to a close, it is important once again to acknowledge some limitations encountered during the research process. The decision to narrow the scope to the top fifteen skaters, while necessary for data verification, may have excluded instances of bias occurring in the lower rankings where judges might feel less watched or scrutinized by popular media. Further, the transition from random drawing and anonymization to the current, more transparent system represented an overwhelming shift in data availability. While my initial dataset contained nearly twenty-six thousand rows of data, the one I ended up utilizing the most only contained a

little more than eight thousand. More data might have helped provide more nuanced takes on the network in Figure 6. It is especially interesting to ponder about the geopolitical blocs.

This project came about through the asking of whether national judging bias is truly a systemic reality or purely a sensationalized media narrative. Through tiered visualization of 8 years of World Championship protocols, the answer is clearly that the data as I see it supports the outrage. From the micro level bumps in isolated instances, to more mega and macro level long term longitudinal trends, it is evident that the human aspect of the IJS is still volatile. But, importantly, this paper also demonstrates the power of data transparency. I was able to reverse-engineer the protocols and gather data to visualize the bias. As such, I am able to hold the system accountable in a way that was impossible under the old 6.0 system, and even the early days of the IJS.

Most recently, at the 2026 Winter Olympic Games in Milano-Cortina, public outcry over the scoring of Chock and Bates versus F-Beaudry and Cizeron highlighted that these tensions might be even more alive today than they were in 2002. However, as a general trend, the data suggests that the IJS, specifically the trimmed mean, remains still largely effective at correcting for national bias. Moving forward, I believe that the ISU should focus on ensuring that the safety net judging can be as transparent as possible, if not also finding a way to minimize its widespread nature.